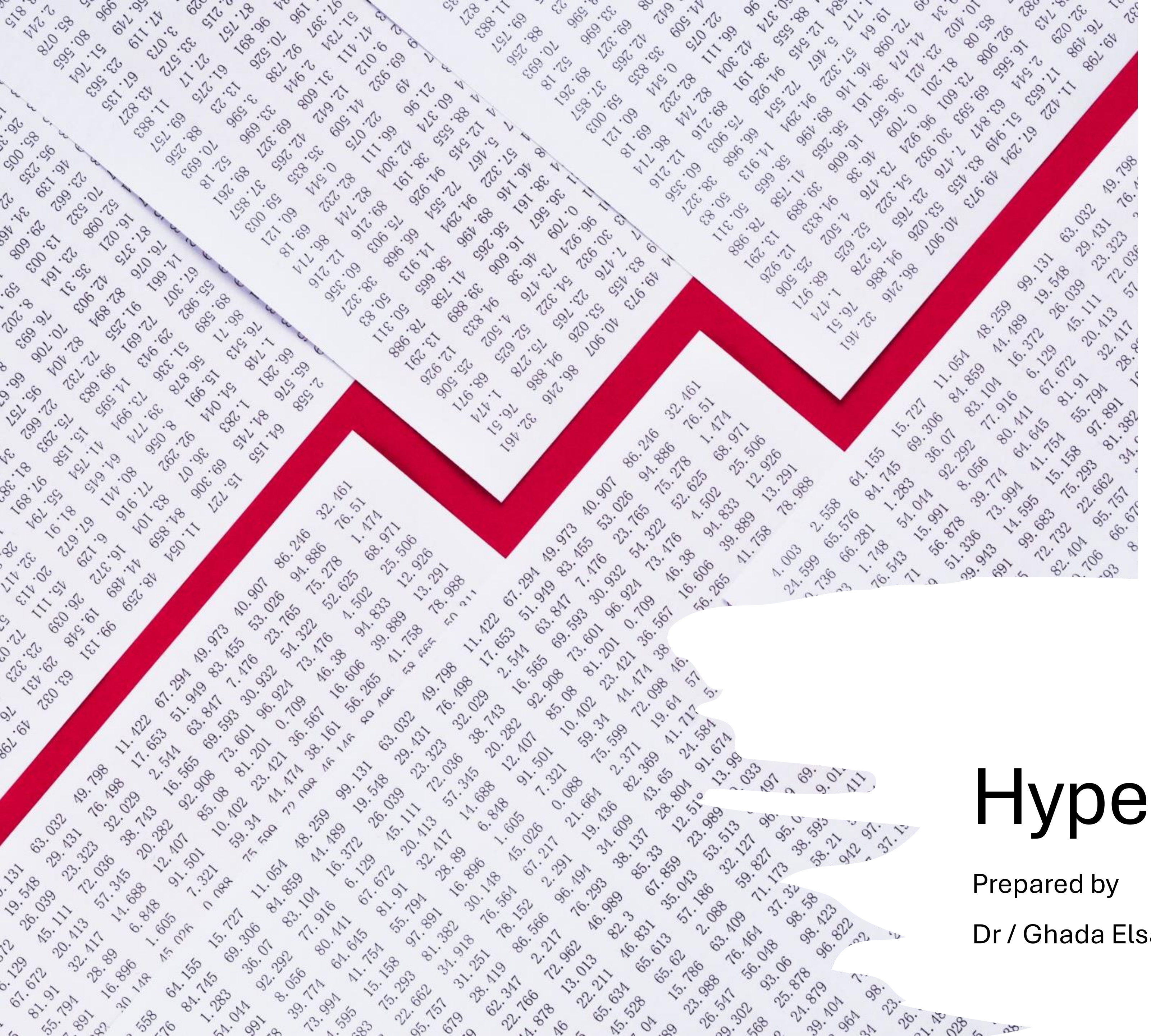

Hypertensive Crisis

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Hypertensive Crisis

Outline

- Definition
- Types of hypertensive Crisis
- Pathophysiology
- Clinical manifestation of emergency crises
- Treatment of emergency crises

Pre test

Mention causes of hypertensin

Mention Types of hypertensive Crisis

*Introduction

- Primary hypertension may occur as an unknown cause.
- However there are several risk factors which predispose a person to hypertension.
- Hypertension is usually diagnosed in men between 30-50 years of age.
- Obesity is one factor as well as smoking, excessive sodium intake, elevated serum lipids and alcohol intake.
- Stress may result in increased in sympathetic nervous system activity.
- In all of these cases hypertension and hypertensive crisis can be prevented or controlled.

Hypertensive crisis

- Hypertensive crisis (emergencies and urgencies)
- occur in less than 1% of the 60 million people with hypertension. Even though not many people with hypertension experience hypertensive crisis, there are still a significant number of events.
- If left untreated, over time hypertension will cause changes in the vessels, resulting in stroke, blindness and renal failure

*Definition: -

Hypertensive crisis is a sever increase in blood pressure that can lead to a stroke ,extremely high blood pressure , greater than 180/110 mm Hg

*Types of hypertensive Crisis:

1. Hypertensive emergency.

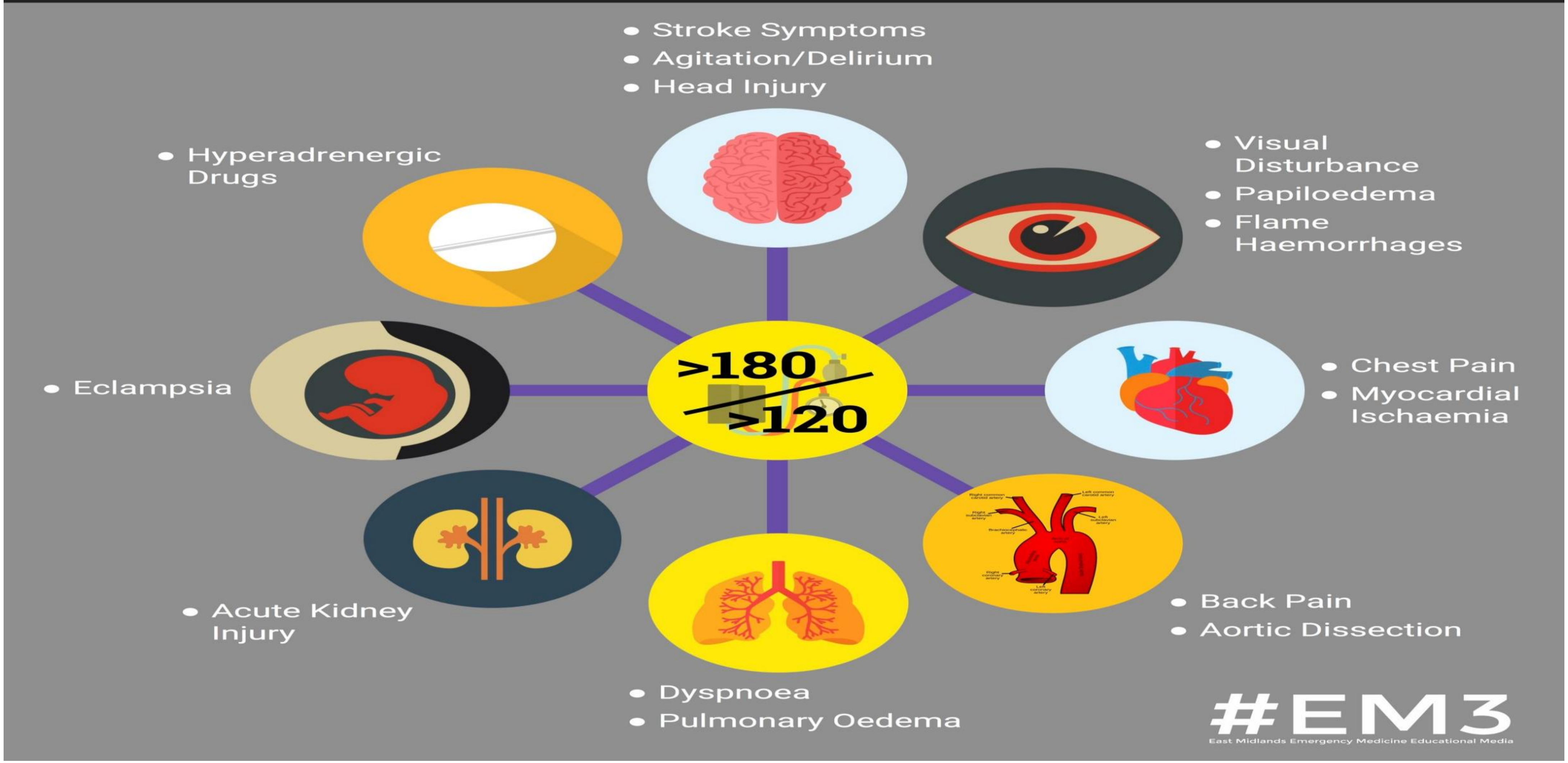
2. Hypertensive urgency

Pathophysiology:

Causes of hypertensive crisis

- Abnormal renal function
- Hypertensive encephalopathy
- Intracerebral hemorrhage
- Heart failure
- Withdrawal of antihypertensive drugs (abrupt)
- Myocardial ischemia
- Eclampsia
- Pheochromocytorma

Hypertensive Emergencies:



Prolonged hypertension Inflammation and necrosis of arterioles Narrowing of blood vessels Restriction of blood flow to major organs

- **Organ damage**

- Renal
- Decreased renal perfusion
- Progressive deterioration of nephrons
- Decreased ability to filtration of body fluid
- Cerebral
- Decreased cerebral perfusion
- Increases stress on vessel wall

• Cardiac

- Decreased cardiac perfusion
- Coronary artery disease
- Angina or myocardial infarction
- Increase cardiac work load
- Left ventricular hypertrophy

1. Hypertensive emergency:-

- **Definition of hypertensive emergency:-**
- It is a sudden increase in systolic and diastolic Bp more than 180/120 mmHg associated with actual or developing dysfunction of target organ.
- It requires prompt treatment in an intensive care setting because of the serious target organ damage that may occur as the CNS, the heart, or the kidney.
- **Conditions associated with hypertensive emergency include:-**
- Hypertension of pregnancy
- Acute MI,
- dissecting aortic aneurysm and intracranial hemorrhage

Clinical manifestation of emergency crises:-

- **1. CNS manifestations:**
- Classic manifestation: -
- headache, altered level of consciousness less severe degree of CNS dysfunction.
- Advanced manifestation: - retinopathy with arterioles changes hemorrhage and exudates and papilledema.
- **2. Cardiovascular manifestations:**
- • Angina or acute myocardial infarction.

- **3- Respiratory Manifestation**:
- Dyspnea.
- Coughs.
- Fatigue.
- Frank pulmonary edema.

4. Renal manifestation:

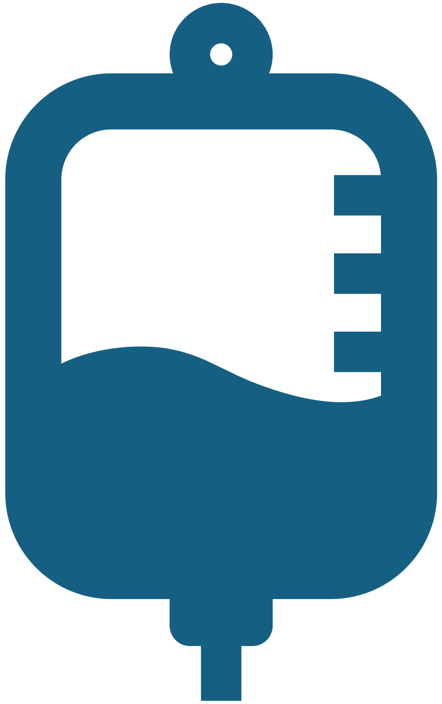
- Renal failure with oliguria and hematuria.

Treatment of emergency crises

- The therapeutic goals are reduction of the mean blood pressure by up to 25% within the first hour of treatment,
- a further reduction to a goal pressure of about 160/100 mmHg over a period of up to 6 hours, and then amore gradual reduction in pressure over a period of days.
- The exception of these goals is the treatment of ischemic stroke and treatment of aortic dissection (in which the goal is to lower systolic pressure to less than 100 mm Hg if patient can tolerate the reduction).

<u>The medication of choice in hypertensive emergencies with an immediate effect:-</u>

- <u>1. Intravenous vasodilators including the following: -</u>
- Sodium nitroprusside (Nipridem, Nitropress), it is arteriolar and venous vasodilator.
- Diazoxide, arteriolar vasodilator.
- Hydralazine, arteriolar vasodilator.



- Labetalol, nonselective beta blocker.
- Nicardipine hydrochloride (Cardene).
- Nitroglycerin has immediate actions that are short lived (minutes to 4 hours) and they are used for initial treatment.

2. Oral medications including the following: -

- Nifedipine, calcium-channel blocker.
- Captopril, angiotensin-converting enzyme inhibitor.
- Minoxidil, arteriolar vasodilator.

2. Hypertensive urgency:-

- **Definition of hypertensive urgency:-**
- Severely elevated BP without acute end organ damage, the clinical differentiation between them depends on the presence of target organ damage rather than level of Bp.
- **Clinical manifestation of urgent crises:-**
- 1. Severe headache, nose bleeds.
- 2. Severe anxiety.
- 3. Shortness of breath.

Treatment of urgency crises:-

- **Oral dose of fast acting agents such as:**
- Beta adrenergic blocking agents (e.g., labetalol or normodyne).
- Angiotensin converting enzyme inhibitors as captopril (capoten) • Alpha 2 agonists as clonidine or catapres are recommended for treatment of hypertensive urgency.

* Management

- **Initial evaluation and investigations of the patient with hypertensive crisis:**
- The key to successful management of patients with severely elevated BP is to differentiate hypertensive emergency from hypertensive urgencies.
- 1- The BP in all limbs should be measured by the physician.
- In obese patients appropriately sized cuffs should be used.
- 2- Funduscopic examination to detect the presence of papilledema.
- 3- CBC count, electrolytes, BUN, creatinine, and urine analysis should be assessed.
- 4- Peripheral blood smears to detect the presence of a microangiopathic hemolytic anemia

- A chest radiography,
- ECG,
- CT scan are useful in patients with evidence of shortness of breath, chest pain, or neurological changes.
- 6- Echocardiogram to assess left ventricular function and evidence of ventricular hypertrophy.

Important aspect of history and physical examination

- History:-
  - Duration of the hypertension.
  - Last known blood pressure.
  - Course of blood pressure.
- Prior treatment of hypertension(types, does , sides effect)
- Intake of agent that may cause hypertension ,

  such as :-
  - Oral contraception, adrenal steroid, and excessive sodium intake.
- Family history Hypertension Cardiovascular diseases Renal disease and DM
- Symptoms of target organ damage :-

Physical examination:-

- Accurate measurement of blood pressure.
- General appearance include disturbance of the body fat, skin lesion, muscle strength and alertness.
- • Funduscopy.
- Neck: -
- palpation and auscultation of the carotid and thyroid area.
- Heart: - Detection of the size of the heart, sound and rhythm.
- • Lung: - Inspection and auscultation of the chest.
- • Abdomen: - To detect any renal masses and femoral pulse.
- • Neurological assessment. • Feature of endocrine abnormalities.

Nursing management:

- ❖ Nursing assessment : The assessment of the client with hypertension has three main objectives:
- i. To determine the extent of target organ damage.
- ii. To ascertain the presence of other cardiovascular disorder.
- iii. To identify the types of antihypertension drugs.
- Note:- All this objectives can be established through history, physical examination and laboratory studies.

- **the following point when you meet hypertensive client:**
- ❖ Family history of hypertension, DM, and any cardiovascular disease.
- ❖ Previous documentation of high blood pressure including age of onset and prescribed medication.
- ❖ History of any diseases or trauma to target organ.
- ❖ Results and sides effect of the antihypertensive drugs used for the patient.

- ❖ Clinical manifestation of the cardiovascular disorder including angina, dyspnea and claudication.
- ❖ History of weigh gain, exercise and dietary habits.
- ❖ Presence of other cardiovascular risk factors including obesity, smoking and alcoholism.
- ❖ Physical examination and laboratory studies are mentioned before

Nursing diagnosis:

- **1. Ineffective management of therapeutic regimen.**
- ❖ Assess the pt understanding to his new condition including:
- a. Description of hypertension and the known risk factor.
- b. The importances of medical follow up.
- c. Listing the prescribed medication.
- ❖ The client with hypertension need clear practical and realistic guidelines about information concerning hypertension and its treatment and follow up.

Simplify the drug regimen including:

- a. The pt planning for treatment.
- b. The benefit of adherence to prescribed regimen.
- c. The negative side effect of prescribed drugs and how it can be controlled.
- d. Instruct and educate the client how to take his blood pressure.
- e. Involve the pts significance other and family in the programs of education and teaching

2- Altered in health maintenance related to knowledge deficit about the disease process.

- ❖ Assess the client knowledge base about the diseases process. ❖ Encourage the question about the hypertension and prescribed medication.
- ❖ Involve the family and significance other.

- ❖ Providing information in following area:

- Definition of hypertension and differentiation between the systolic and diastolic pressure.
- b. Causes of hypertension.
- c. Nature of the disease and its target organ damage.
- d. Treatment goal being to control versus cure.
- e. Rational and strategies of weigh reduction.
- f. Avoidance of coffee, tea, cola and chocolate.

- **3. Activity intolerance related to imbalance between the oxygen supply and demand secondary to decreased cardiac out put.**
- ❖ Assess activity tolerance and current ability of pt to conduct activities of daily living.
- ❖ Space nursing activities.
- ❖ Schedule rest period.
- ❖ Monitor the client response to activities.
- ❖ Increased activities as ordered or according to rehabilitation nursing activities.
- ❖ Instruct the client to avoid the activities that increased cardiac workload.

- Monitor cardiopulmonary signs of activity intolerance (tachycardia, hypotension, and tachypnea).
- ❖ Prolonged bed rest and self imposed bed rest must be avoided to prevent its hazard complication such as bed sore and pressure ulcer especially in edematous pt and to prevent phlebothrombosis and pulmonary embolism.
- ❖ Vital signs must be taken before, after and during activity.
- ❖ During the acute stage the nurse provide the self care to the pt until to be able to participate in it.

4-Fluid volume excess related to reduce glomular filtration, decreased cardiac out put, increased antidiuretic hormone production and sodium, water retention.

- ❖ Monitor intake and out put every 4 hours more or less frequently depends on the severity of case.
- ❖ Weight client daily.
- ❖ Asses for jugular vein distension, hepatomegally and abdominal pain.
- ❖ Assess for peripheral edema.
- ❖ Provide low sodium diet and fluid restriction

- Auscultate breath sound to detect any presence of crackle.
- ❖ The nurse needs to teach the pt to assume the position that shift the fluid away from the heart by elevation of head of the bed by billow or by mechanical method.
- ❖ Provide oral care to pt frequently especially whose take their breath from the mouth.
- ❖ Assess and monitor signs of increasing peripheral edema.
- ❖ Assess renal function through blood analysis for urea and creatinine.

5-Altered peripheral, cerebral and renal tissue perfusion related to decreased cardiac out put.

- ❖ Note color and temperature of the skin.
- ❖ Monitor peripheral pulse.

Provide a warm environment to provide vasodilatation.

- ❖ Encourage active rang of motion.
- ❖ Monitor urine out put.
- ❖ Early detection of sign of thrombophlebitis.
- ❖ Observe mental status constantly.
- ❖ Monitor fluid intake and out put.
- ❖ Maintain adequate blood pressure.
- ❖ Avoid inducing hypotension

* Follow –up:-

- The pt should be monitored every 1 to 2 week until the target blood pressure is achieved, then every month for 3 month then every 6 month indefinitely, unless the blood pressure become unstable.
- creatinine and potassium level and urine analysis should be evaluated annually as part of screening.

Teaching Patient about the following:

- 1. Weight control :-
- Overweight people have two to three times the risk for development of hypertension than people who are not overweight.
- 2. Physical activity:-
- The results of the study reveal that the physical activities reduce the elevation of blood pressure which is combined with weight loss
- . 3. Reduction of alcohol intake and smoking:-
- The data from epidemiological study indicate that alcohol intake associated with increased in blood pressure.
- And increased intake of the alcohol increased both

Client education toward low sodium diet:

- ❖ Avoid food high in sodium such as:
- ❖ Avoid the common preparation that is high in sodium, including backing soda monosodium glutamates, meat tenderizer and soy sauce.
- ❖ Avoid canned food and some frozen food in which sodium has been added.
- ❖ Avoid canned smoked food or cured meat and fish products

Modification of dietary fat:-

- Content 159 By decreasing saturated fat intake and increasing poly- unsaturated fat, it will decrease cholesterol level which is important risk factor for coronary heart diseases.
- The use of fish oil supplements to lower cardiovascular risk seen as the reducer of blood pressure as it demonstrated in the study.

Client education about low fat in diet:

- ❖ Avoid food high in saturated fats and cholesterol:
- ❖ Use margarine and vegetables oil instead of butter.
- ❖ Avoid graves, creams and cheese sauces.
- ❖ Use skim or low fate milk and milk products.
- ❖ Choose lean cut of meat, trim off all visible fat.
- ❖ Prepare meat stew, soup and graves in advance and chill them until fat become harden.
- ❖ Eat no more then three egged yolk per week, egg white is low in cholesterol. ❖ Limit your intake of organ meat and shellfish.

- **6. Relaxation technique:-**
- Varity of relaxation therapies including yoga and psychotherapy has been shown o reduce blood pressure.
- **7. Potassium supplementation: -**
- The high ratio of sodium to potassium in the modern diet has been held responsible for the development of hypertension, so the potassium supplement may lower the blood pressure.

Post test

- Mention risk factors of hypertensive crisis
- Mention the effect of hypertensive crisis on the body organs damage

Any question

Thank you